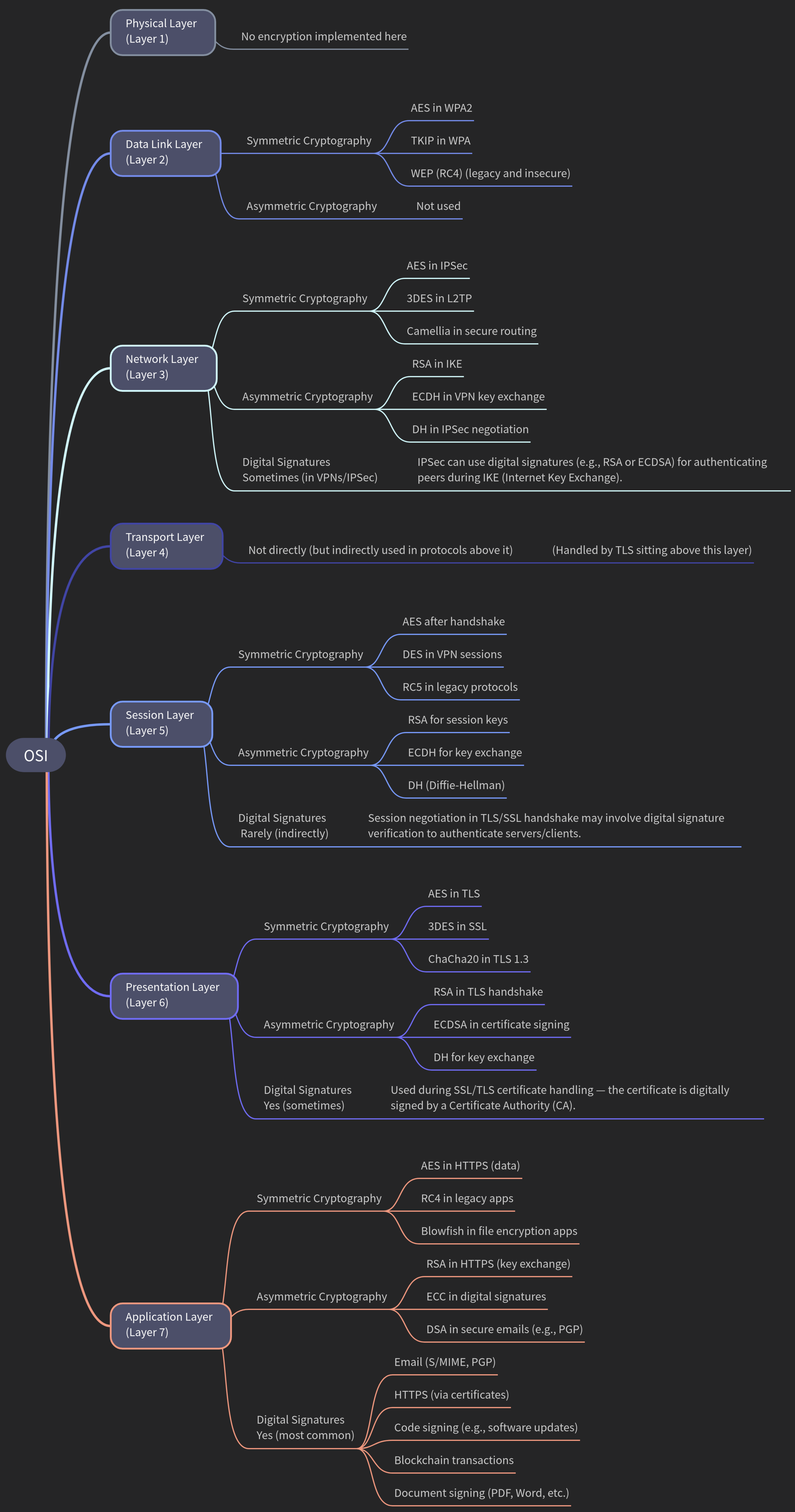
OSI
Physical Layer (Layer 1)
No encryption implemented here
Data Link Layer (Layer 2)
Symmetric Cryptography
AES in WPA2
TKIP in WPA
WEP (RC4) (legacy and insecure)
Asymmetric Cryptography
Not used
Network Layer (Layer 3)
Symmetric Cryptography
AES in IPSec
3DES in L2TP
Camellia in secure routing
Asymmetric Cryptography
RSA in IKE
ECDH in VPN key exchange
DH in IPSec negotiation
Digital Signatures Sometimes (in VPNs/IPSec)
IPSec can use digital signatures (e.g., RSA or ECDSA) for authenticating peers during IKE (Internet Key Exchange).
Transport Layer (Layer 4)
Not directly (but indirectly used in protocols above it)
(Handled by TLS sitting above this layer)
Session Layer (Layer 5)
Symmetric Cryptography
AES after handshake
DES in VPN sessions
RC5 in legacy protocols
Asymmetric Cryptography
RSA for session keys
ECDH for key exchange
DH (Diffie-Hellman)
Digital Signatures Rarely (indirectly)
Session negotiation in TLS/SSL handshake may involve digital signature verification to authenticate servers/clients.
Presentation Layer (Layer 6)
Symmetric Cryptography
AES in TLS
3DES in SSL
ChaCha20 in TLS 1.3
Asymmetric Cryptography
RSA in TLS handshake
ECDSA in certificate signing
DH for key exchange
Digital Signatures Yes (sometimes)
Used during SSL/TLS certificate handling — the certificate is digitally signed by a Certificate Authority (CA).
Application Layer (Layer 7)
Symmetric Cryptography
AES in HTTPS (data)
RC4 in legacy apps
Blowfish in file encryption apps
Asymmetric Cryptography
RSA in HTTPS (key exchange)
ECC in digital signatures
DSA in secure emails (e.g., PGP)
Digital Signatures Yes (most common)
Email (S/MIME, PGP)
HTTPS (via certificates)
Code signing (e.g., software updates)
Blockchain transactions
Document signing (PDF, Word, etc.)